

t≡rrΛnoVΛ

Surface that hold a room together.

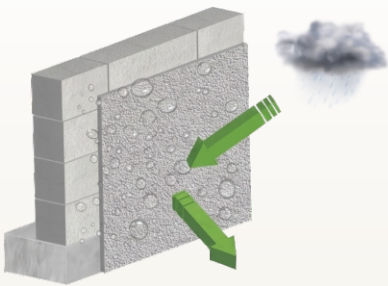


HDS HYDROBLOC DRYING SYSTEM

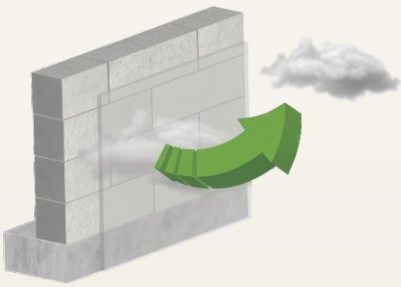
Unique category of products for building's finishing purpose based on European technology (Slovakia) called **Terraonv**. As first system in the world, HDS combines 100% waterproof ability with extreme breathability and moisture removal ability called drying process, all implemented into plastering mortars.



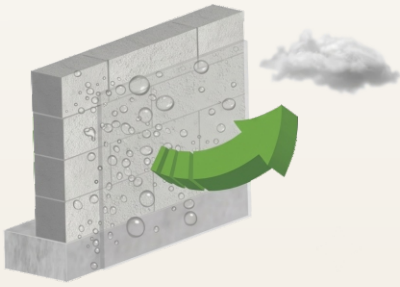
100% Plaster



100% Water Proof

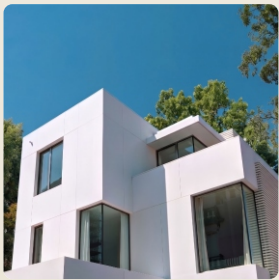


100% Breathable



100% Drying Process

HDS HYDROBLOC DRYING SYSTEM



WATERPROOF MEMBRANE





HDS PLASTER

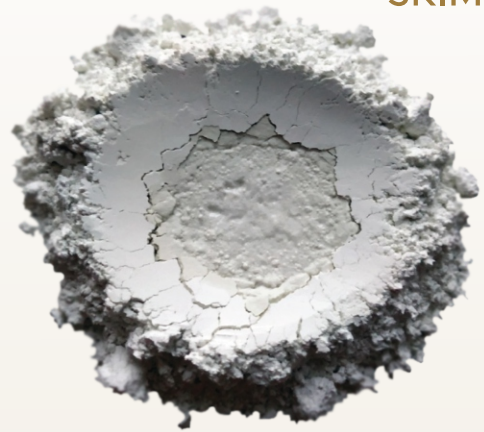


PLASTER

HDS SKIM
COAT



SKIM COAT



HDS mixing



HDS mixing



HDS application



HDS application



HDS application



HDS Plaster

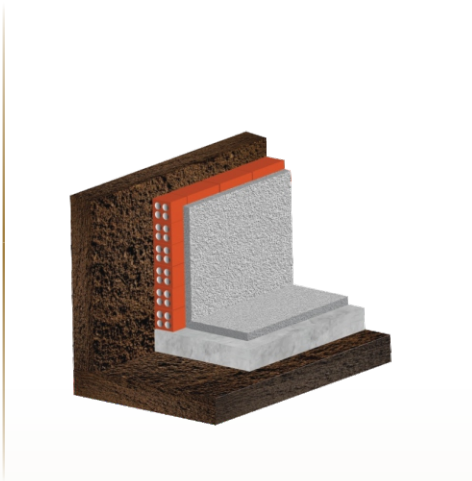
Plaster



HDS Plaster

Plaster

HDS - HARD | 25KG



TYPE	Ready to use product - gray powder
WATER RATIO	3,7-4,0 Liters per 25 kg of material
MIXING	5 minutes with electric hand mixer, 5 minutes rest, 1 minute mixing again 10 minutes in gravity (concrete) mixer
SUBSTRATE	Minimum 10 mm (wall), minimum 20 mm (floor)
COVERAGE	2 m ² @ 10 mm, 1 m ² @ 20 mm
PENETRATION	≤ 1 mm under 1 bar (0,1 MPa) positive or negative water pressure
USE	Basements Water tanks Swimming pools Bathrooms, roof decks, balconies terraces
NOTE	For optimal performance, fiberglass mesh is always used Specification of fiberglass mesh: 5 mm x 5 mm (size) 145 g/m ² (weight) Coated with SRB glue (Type A)

HDS - HARD - EN 988 - 1

Table No. 1: Test results and evaluation of the product TERRANOVA HARD

Test parameter	Test method	Prescription for conformity attestation	Measured value	Evaluation
Flow of fresh mortar after mixing [mm]	STN EN 1015-3	Informative value	160	-
Workable life of fresh mortar [min]	STN EN 1015-9, method B	TL ≥ 180	345	Meets
Dry bulk density of hardened mortar [kg/m ³]	STN EN 1015-10	Declared value TL 1600 ± 100	1580	Meets
Compressive strength after 28 days [N.mm ²]	STN EN 1015-11	STN EN 998-1 tab. 2 CS II (from 1,5 to 5,0) TS ≥ 3,40	12,3	Meets
Flexural strength after 28 days [N.mm ²]	STN EN 1015-11	Informative value TL ≥ 1,6	3,0	-
Adhesive strength of hardened mortar on substrate [N.mm ²] and failure pattern FP a, b or c)	STN EN 1015-12	Informative value TL ≥ 0,3	0,5 – FP a)*)	Meets
Capillarity (coefficient of capillary absorption) [kg/m ² .min ^{0.5}]	STN EN 1015-18	STN EN 998-1 tab. 2 ≥ 0,3 after 24 h	1,6	Meets
Water penetration depth after the test of capillarity [mm]	STN EN 1015-18	STN EN 998-1 tab. 2 ≤ 5 TL < 1	0	Meets
Thermal conductivity coefficient [W/m.K]	STN EN 1745 tab. A.12	STN EN 998-1 tab. 2	≤ 0,45 table value	-
Water vapour permeability coefficient μ [-]	STN EN 1015-19	STN EN 998-1 tab. 2 ≤ 15	3	Meets
Frost resistance coefficient after 10 cycles freeze thaw [-]	STN 72 2452	STN 72 2452 min. 0,75	1,07	Meets
Reaction to fire	Without testing	Comission delegated regulation (EU) 2017/1228	A2 – s1, d0	-
Release of dangerous substances to environment	Checking of SIC	Annex II of Regulation (EC) No. 1907/2006/EC	Safety information card	-
Weight activity of ²²⁶ Ra (Bq. kg ⁻¹)	Gamma spectrometry analyze	Regulation of MZ SR No. 528/2007 coll max. 120	9,0 ± 1,6	Meets
Index of weight activity of radionuclides (-)	Gamma spectrometry analyze	Regulation of MZ SR No. 528/2007 coll max. 1	0,15 ± 0,02	Meets

Note: *) Pattern of failure of adhesion - failure of mortar and substrate. Value from test is equal to the value of adhesion

HDS - HARD - EN 12390-8

- Water pressure 1 BAR (0.1 MPa)
- Test duration 72 hours

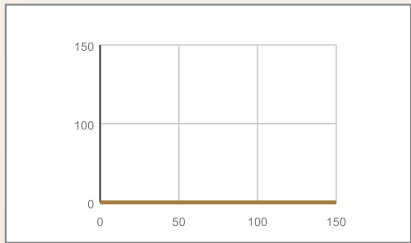
TEST RESULTS:

Age of samples at the beginning of testing: 31 days Direction of water pressure action :
Maximal water pressure: 0,5 Mpa upwards, against the line of mould filling
Water leakage to surface: None

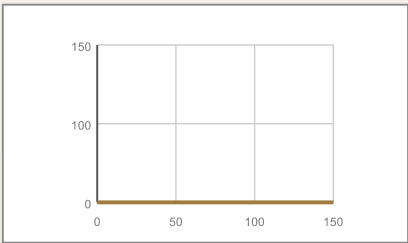
Number of sample:		1	2	3
Weight of sample:	(kg)	4,72	4,69	4,68
Height of tested sample:	(mm)	150,9	150,6	151,1
Width of tested sample:	(mm)	149,9	150,0	149,8
Length of tested sample:	(mm)	150,1	149,6	149,9
Volume density of concrete:	(kg.m ⁻³)	1390	1390	1380
Defined maximal values of leakage on sample:	(mm)	0	0	0
Defined average values of leakage on sample:	(mm)	0		

Graphic presentation:

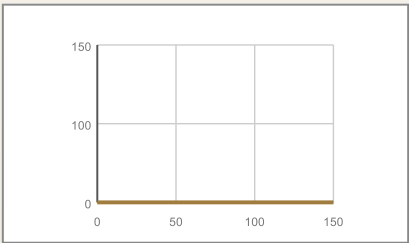
sample No. 1:



sample No 2:



sample No. 3:

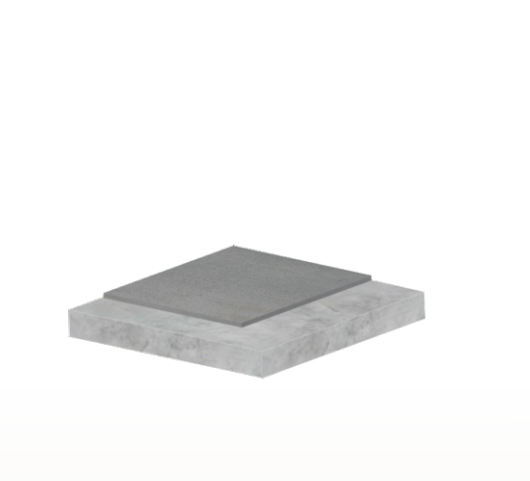


Discrepancy to unified method: the samples were exposed to water pressure 0,1 Mpa!

Test evaluation:

*By testing of water permeability were determined the maximal leakages of 0, 0, 0 mm.
By testing of water permeability were determined the average leakages of 0 mm.*

HDS - HARD | 25KG



TYPE	Ready to use product - gray powder
WATER RATIO	6,2-6,5 Liters per 25 kg of material
MIXING	5 minutes with electric hand mixer, 5 minutes rest, 1 minute mixing again 10 minutes in gravity (concrete) mixer
SUBSTRATE	Smooth (casted) concrete wall, concrete slab
THICKNESS	Minimum 5 mm (wall), minimum 5 mm (floor)
COVERAGE	25 kg = 3,5 m² @ 5 mm, 2,5 m² @ 7 mm
PENETRATION	≤ 1 mm under 1 bar (0,1 MPa) positive or negative water pressure
USE	Basements Water tanks Swimming pools Bathrooms, roof decks, balconies terraces
NOTE	For optimal performance, fiberglass mesh is always used Specification of fiberglass mesh: 5 mm x 5 mm (size) 145 g/m² (weight) Coated with SRB glue (Type A)

Table No. 1: Test results and evaluation of the product TERRANOVA HARD

Test parameter	Test method	Prescription for conformity attestation	Measured value	Evaluation
Flow of fresh mortar after mixing [mm]	STN EN 1015-3	Informative value	167	-
Workable life of fresh mortar [min]	STN EN 1015-9, method B	TL ≥ 180	225	Meets
Dry bulk density of hardened mortar [kg/m ³]	STN EN 1015-10	Declared value TL 1600 ± 100	1390	Meets
Compressive strength after 28 days [N/mm ²]	STN EN 1015-11	STN EN 998-1 tab. 2 CS II (from 1,5 to 5,0) TS ≥ 20,0	22,5	Meets
Flexural strength after 28 days [N/mm ²]	STN EN 1015-11	Informative value TL ≥ 3,0	3,0	-
Adhesive strength of hardened mortar on substrate [N/mm ²] and failure pattern FP a, b or c)	STN EN 1015-12	Informative value TL ≥ 1,5	1,6 – FP a)*)	Meets
Capillarity (coefficient of capillary absorption) [kg/m ² .min ^{0,5}]	STN EN 1015-18	STN EN 998-1 tab. 2 ≥ 0,3 after 24 h	1,7	Meets
Water penetration depth after the test of capillarity [mm]	STN EN 1015-18	STN EN 998-1 tab. 2 ≤ 5 TL < 1	1	Meets
Water vapour permeability coefficient μ [-]	STN EN 1015-19	STN EN 998-1 tab. 2 ≤ 15	14	Meets
Frost resistance coefficient after 10 cycles freeze thaw [-]	STN 72 2452	STN 72 2452 min. 0,75	1,07	Meets
Reaction to fire	Without testing	Comision delegated regulation (EU) 2017/1228	A2 – s1, d0	-
Release of dangerous substances to environment	Checking of SIC	Annex II of Regulation (EC) No. 1907/2006/EC	Safety information card	-
Weight activity of ²²⁶ Ra (Bq. kg ⁻¹)	Gamma spectrometry analyze	Regulation of MZ SR No. 528/2007 coll max. 120	9,0 ± 1,6	Meets
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Note: *) Pattern of failure of adhesion – failure of mortar and substrate. Value from test is equal to the value of adhesion

HDS - HARD - EN 12390-8

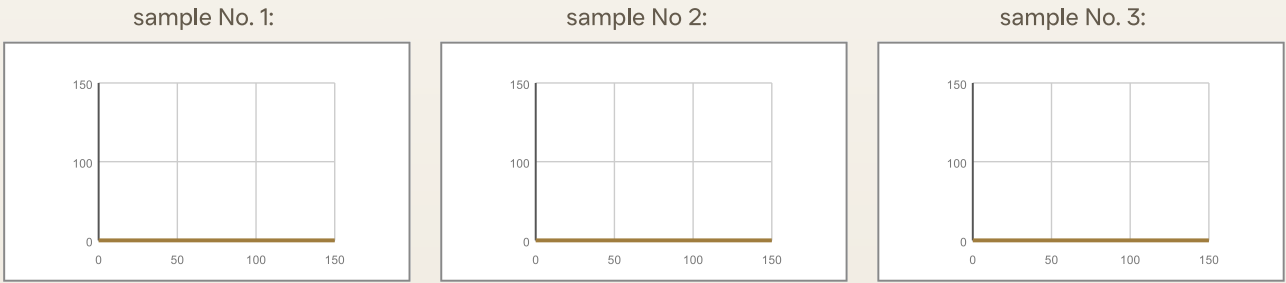
- Water pressure 1 BAR (0.1 MPa)
- Test duration 72 hours

TEST RESULTS:

Age of samples at the beginning of testing:	34 days	Direction of water pressure action :
Maximal water pressure:	0,5 Mpa	upwards, against the line of mould filling
Water leakage to surface:	None	

Number of sample:		1	2	3
Weight of sample:	(kg)	6,23	4,69	4,68
Height of tested sample:	(mm)	150,5	151,5	151,2
Width of tested sample:	(mm)	149,6	150,2	150,2
Length of tested sample:	(mm)	150,3	149,8	149,9
Volume density of concrete:	(kg.m ⁻³)	1840	1850	1850
Defined maximal values of leakage on sample:	(mm)	0	0	0
Defined average values of leakage on sample:	(mm)	0		

Graphic presentation:



Discrepancy to unified method: the samples were exposed to water pressure 0,1 Mpa!

Test evaluation:

*By testing of water permeability were determined the maximal leakages of 0, 0, 0 mm.
By testing of water permeability were determined the average leakages of 0 mm.*

HB - RAIN RESISTANT SYSTEM

PRODUCT DESCRIPTION

Air-entraining and plasticizing natural admixture for plastering mortars to improve workability, to ease application process and to enable application on humid or wet masonry substrate. Plastering mortar modified with HB significantly reduce water abortion of plastering mortar.

PRODUCT USAGE

HB is used for modification of Portland cement / silica sand-based plastering mortar to significantly reduce water abortion suitable for renovation of existing buildings or application at new buildings on external walls. Reduce creation of molds, fungi, wet maps or efflorescence. Lifetime of plastering mortar modified with HB multiple times exceeds the lifetime of general purpose plastering mortars and extend the life span of paint application.

PRODUCT MIXING

Recommended mixing is made in gravity mixer. 19 Litter of water is added to the mixer as well followed by 50 KG cement 30 KG sand and 1 bag (2 KG) of HB. NOTE: Covering the mixer is recommend to reduce dusting. Mixing for 5 minutes then remaining material is added (120 kg sand) and is mixed for additional 5 minutes. Add water if needed. NOTE: HB CAN REDUCE THE WATER REQUIRED TO ACHIEVE DESIRABLE WORKABILITY. APPLICATION WITH FIBERGLASS MESH IS STRONGLY RECOMMENDED

IMPORTANT NOTE

HB can NOT used as an admixture for production of structural concrete.



2 KG

+ 200 kg plaster

50 kg cement / 1 part

150 kg sand / 3 parts

12 m² coverage @ 10 - 15 mm

Can be mixed manually.

Includes plasticizer, plaster

application is very easy

HB

HYDROBLOC SYSTEM

USAGE	MIXING FORMULA
Exterior wall	Exterior wall: 2 kg RRS + 200 kg plaster (200 kg of both cement + sand in ratio 1:3, 1:4 or 1:5) + water
Interior wall Bathroom (wall and floor) Roof deck	Interior wall, Bathroom (wall and floor), Roof deck: 2 kg RRS + 50 kg plaster (50 kg of both cement + sand in ratio 1:3, 1:4 or 1:5) + water

HDS SELECTED REFERENCES



tεrranova

Surface that hold a room together.



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